Can L2 speakers create new categories?

A brief account on category issues in L2 phonology

Fernanda Barrientos
University of Manchester

Why category creation?

Two approaches for this question

- Literally: why is this interesting in general?
- In a more restricted sense: why is this important for phonological theory? Is there another way to explain reasonable communication in L2?

Possible scenarios for L2 speakers

- Category merging (2-to-1)
- **Category splitting (1-to-2)**
- Same (but not quite) (1-to-1)

Splitting categories: issues

- When the L2 has more categories than the L1:
  - Which one is the L1 splitted category?
  - Which one is the L2 category that squeezes in?
  - What happens when the L2 category is in the boundary of more than two L1 categories?

- When / how / why does category splitting take place?
  - Proportional to input (only a matter of quantity? How much is enough?)
  - Age-sensitive?
  - Sensitive to certain types of input / linguistic tasks ?

Alternatives to category creation

Why should we think of an alternative?

- L2 proficiency does not necessarily mean that the speaker has an L2 inventory similar to that of a native speaker.
- Claims from L2 phonology research:
  - No creation of phonological categories (Flege 1995, Best 2007)
  - Strong native language magnet effect after 6mo (Kuhl 1995)
  - Transfer and input-driven reconfiguration of cue-weighting strategies (Escudero 2005)
  - Perceptual assimilation in L2 (Darcy et al 2009)

Then, what?

- Given that
  - … Critical period hypotheses state that humans cannot create new phonemic categories after a certain age
    - ...Which are supported by empirical findings (NLM);
  - … Discrimination tasks in L2 do not show native-like categorization, even when speakers have a high level of proficiency; and
  - … Native language perceptual strategies are initially transferred to L2 perception...

Then, what?

- … Category creation might not be available at all!
  - If not, then what's left?
    - Raw perceptual phenomena
    - Use of L1 knowledge, where mapping is distributed in a 2-to-1 manner, or with possible rearrangement when needed.

Experiment 1: An L2 perceptual map

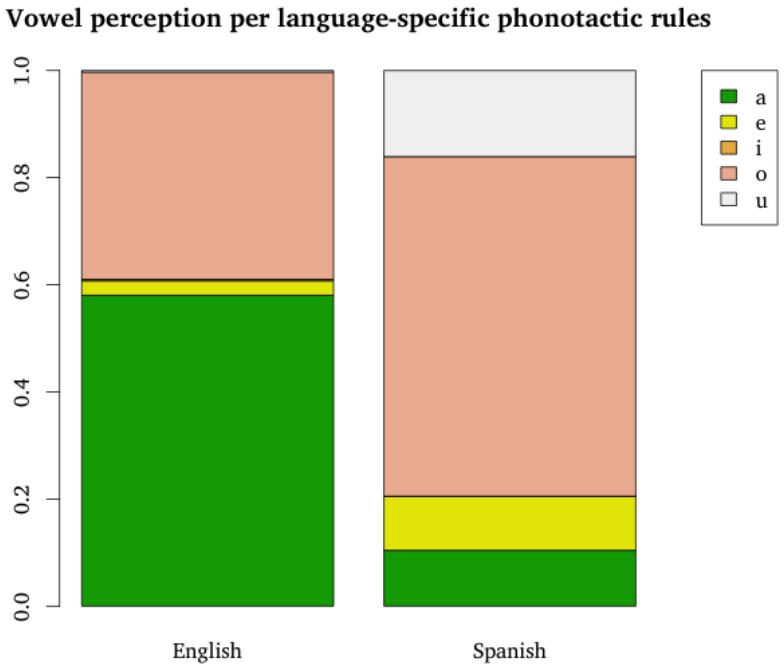
- Are perceptual strategies language-specific?
  - Could phonotactic constraints of a language affect perception of segments?
  - Native speakers of Spanish of different proficiency levels in English categorized English vowels into Spanish categories.
  - English monophthongs were embedded in C_C frames that resulted in nonwords.
    - One group had valid onset-coda combinations in English only.
    - The other group had valid onset-coda combinations only in Spanish.
    - Forced-choice: a, e, i, o, u.

Results

For /ʌ/

Categorized as /o/ in Spanish phonotactic context

- Categorized as /a/ in English context

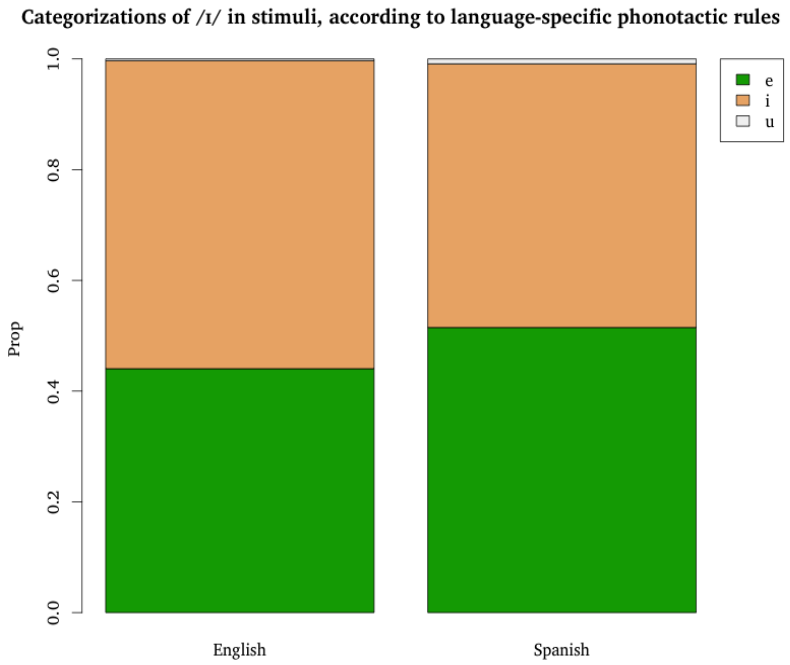


Results

For /ɪ/

Categorized as /e/ in Spanish phonotactic context

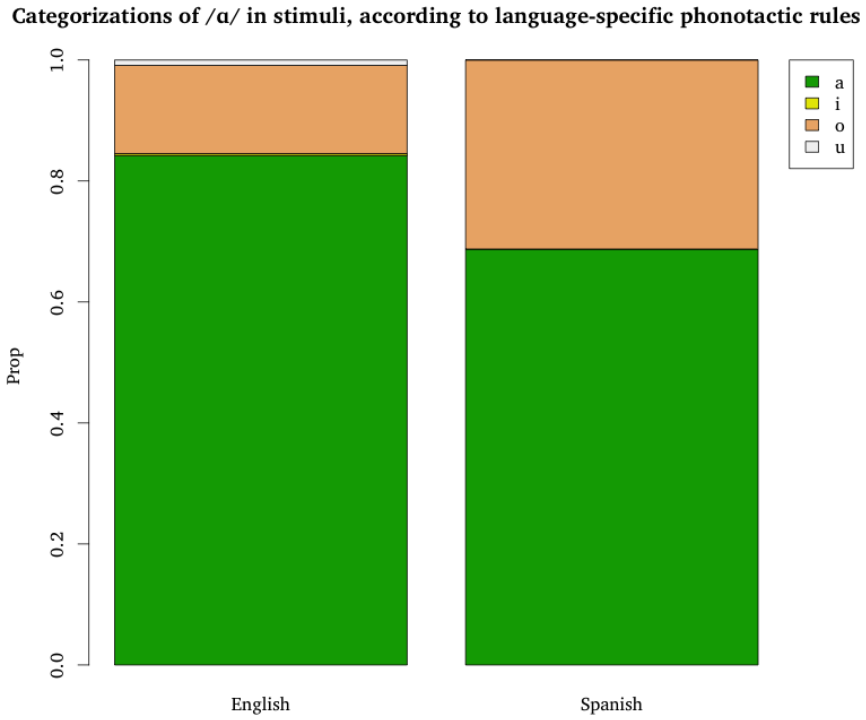
- Categorized as /i/ in English context
- Not statistically significant (.01)



Results

For /ɑ/

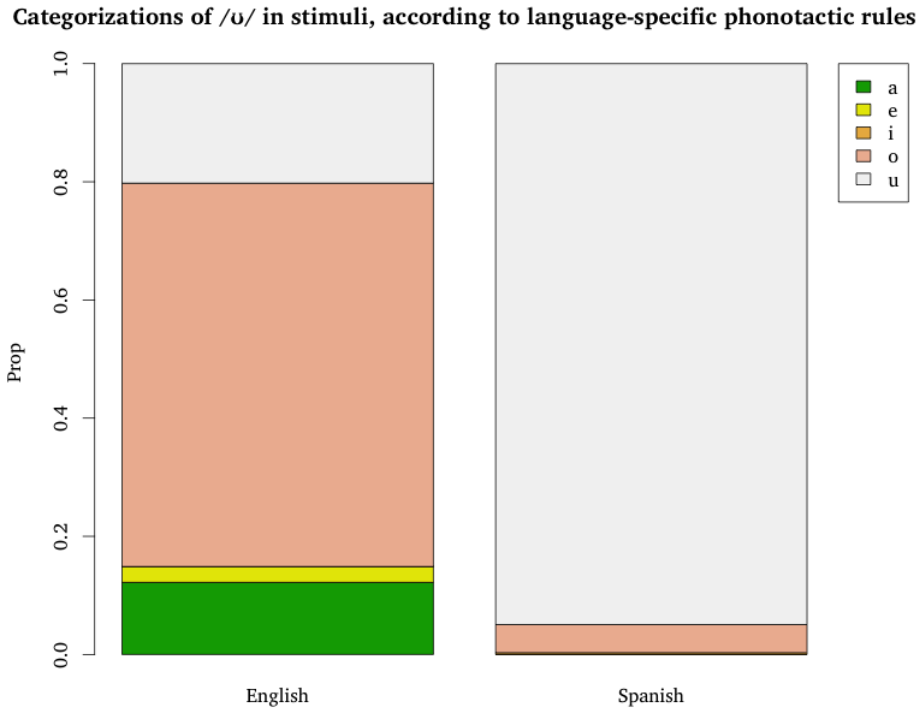
- Categorized as /a/ in both contexts, but even (significantly) more so in English context



Results

For /ʊ/

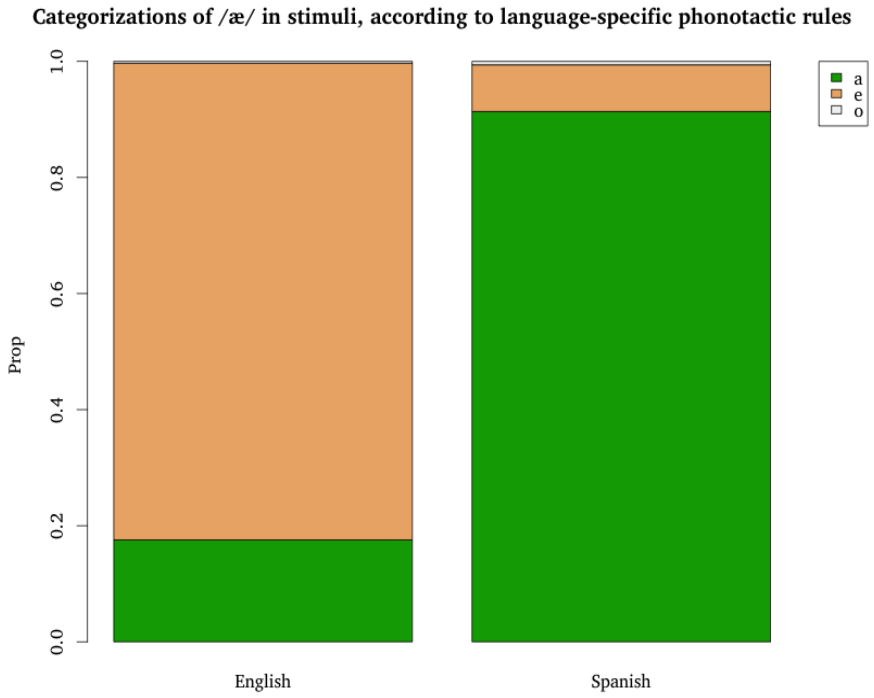
- Categorized as /u/ in Spanish phonotactic context
- Categorized as /o/ in English context



Results

For /æ/

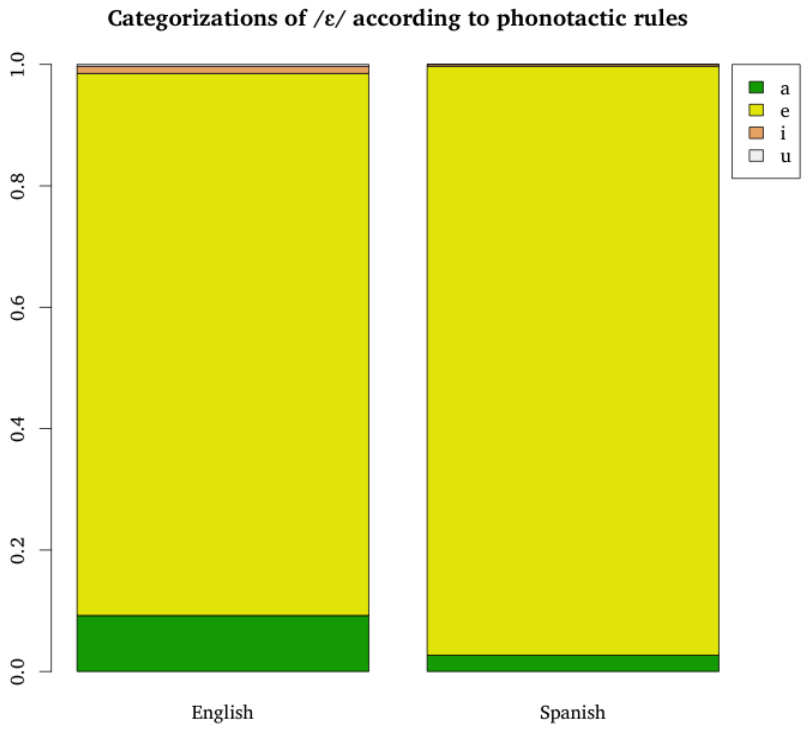
- Categorized as /a/ in Spanish phonotactic context
- Categorized as /e/ in English context



Results

For /ɛ/

- Categorized as /e/ in both contexts
- No statistically significant differences (.01)



Conclusions

- Change in categorizations per vowel
  - Highly significant: STRUT, FOOT, TRAP, LOT
  - Non significant: DRESS, KIT
  - Preference for one formant over another, which triggers categorizations in which the same vowel is perceived as more open/closed or fronted/backed.

Conclusions

- Language-specific perceptual grammars?
  - Changes in cue-weighting
  - F1 y F2 are differently weighted according to perceived phonotactic context

Vowel	English	Spanish
ɪ	more closed	more open
ɑ	more fronted	less fronted
ʌ	more fronted and open	less fronted and more closed
ɛ	more open	more closed
æ	more closed	more open
ʊ	more open	more closed

Experiment 2: Finding boundaries

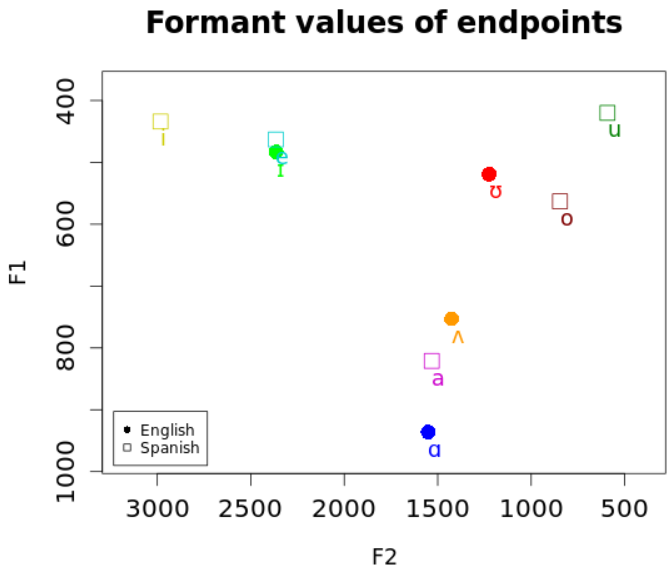
- Can L2 speakers develop categorical perception in the way native speakers do?
- Do they show the same boundaries?
- Will advanced L2 speakers have same boundaries than the inexperienced?

1	2	3	4	5	6	7
/ʌ/						/a/

1	2	3	4	5	6	7
/ɑ/						/o/

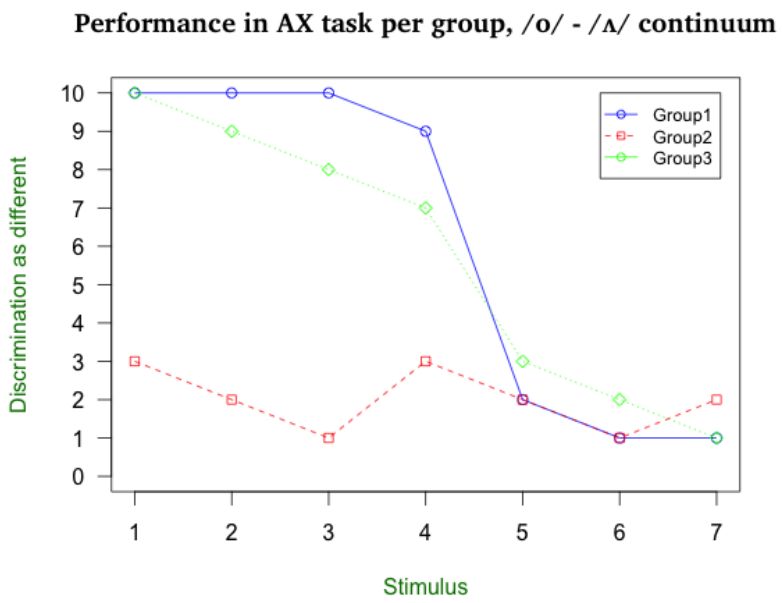
- Construction of a 7-step continuum between a native and an L2 category.
- Will subjects perceive token 4 of the first continuum as same or different than /ʌ/ or /a/?
- Will the boundaries be closer to one end of the continuum than the other? Will it happen in all the continua created? (i.e. will the boundary be between, say, tokens 4 and 5 at all times?)

Expected results



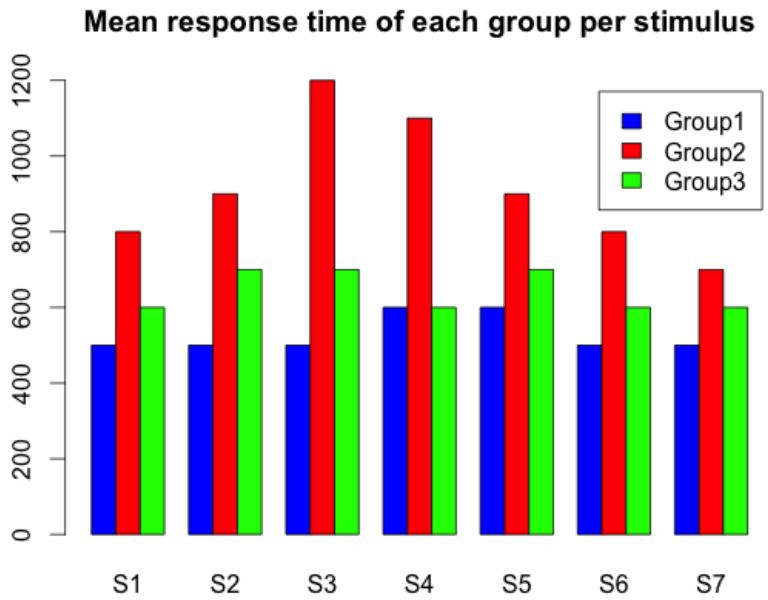
- The closer the endpoints are, the less likely it will be that L2 speakers (of any level) would be able to find differences along the continuum.
- L2 speakers will perceptually assimilate the L2 category to the closest native one in the perceptual space.

Expected results



- Group 1: native speakers (blue curve)
- Group 2: beginners (red curve)
- Group 3: advanced (green curve)
- Groups 1 and 3 should show a similar pattern.

Expected results



- Response time should be higher for Group 2 subjects.
- In the case of groups 1 and 3:
- Shorter response times throughout the continuum, or one sharp increase in boundaries.

...Thoughts?